

Lecture 6: Data Analysis 1: Data Cleaning and Wikipedia

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

Recap: Social Media Data Collection

Social Media Data Collection

- For large-scale, computational studies of social media you need to collect large, representative datasets of social media behavior
 - Ideally, a **'census'**, i.e., the complete set of users and content on the platform
 - But collecting this is very hard without direct access to the platform
 - Alternative: a representative **sample**
- Samples can be collected using different approaches
 - Collection mechanics: Mainly web scraping and APIs (official or unofficial)
 - Sampling strategy: depends on your research question or goal
 - Keywords
 - Snowball sampling
 - Communities
 - External lists
 - ...

APIs vs Web Scraping

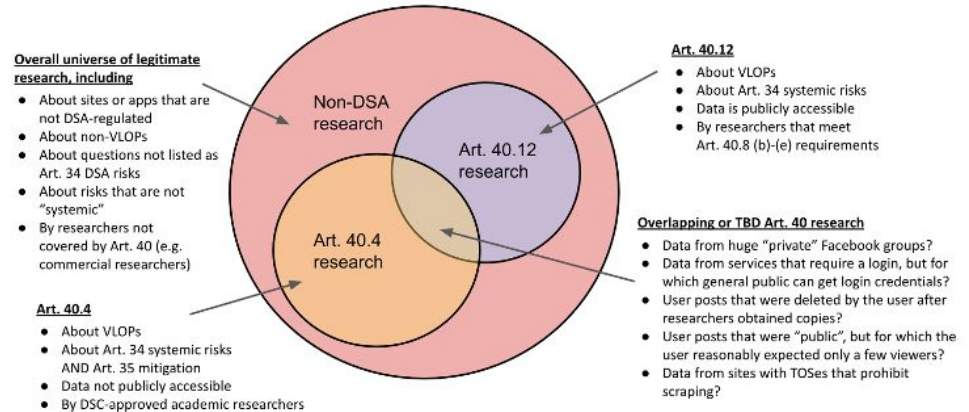
Category	Web Scraping	API-Based Data Collection
Access Method	Extracts data from website front-end (HTML pages)	Sends structured requests to official backend endpoints
Official Support	Not officially supported by platforms	Officially provided by platforms like Facebook, Instagram, X, and LinkedIn* Unofficial APIs also exist but much less stable
Data Format	Unstructured HTML → requires parsing	Structured JSON/XML
Stability	Breaks when website layout changes	Stable and version-controlled
Authentication	May bypass login systems	Requires secure login (OAuth, tokens) [official APIs!]
Permissions	No formal permission model	Uses scoped access permissions
Rate Limits	No official limits, but risk of blocking	Clearly defined request limits
Legal Risk	Potential Terms of Service violations	Governed by developer agreements
Maintenance Effort	High – frequent updates required	Lower – documented and supported
Ethical Considerations	May raise privacy concerns	Designed with privacy safeguards

Platform Affordances, Sampling, and Data Quality

- Platform affect user behavior and the data they generate on these platforms
- Have to be especially careful during sudden affordance changes
 - Are your findings of platform behavior affected by changes?
- On the flipside: changes in the platform are excellent opportunities to study the impact of these affordances
 - Can also be done retroactively
 - See studies in the “resources” section of today’s lecture slides

Data Access

- We discussed the API apocalypse and how many platforms have closed their academic APIs
 - Some platforms have official APIs but they are limited or hard to access (e.g., Instagram, Tiktok)
- However, that doesn't mean researchers stopped studying those platforms
- In fact, regulatory bodies help researchers audit and study these platforms, especially VLOPs (Very Large Online Platforms)
- One such mechanism: the EU's Digital Services Act (DSA)



Daphne Keller:

<https://verfassungsblog.de/dsa-platforms-digital-services-act/>

Do you feel prepared to collect your own
social media data?

Today:

- 1. Data Cleaning**
2. Studying Wikipedia

Now that we have the data, what do we do with it?

- Analyze it! Answer your research questions!
- You can use a wide range of **computational** methods, depending on your goal:
 - Text analysis, Natural Language Processing (NLP), also called “text-as-data” methods: analyze textual features of the data — topics, framing,
 - Network analysis: e.g., find the most central units of your data
 - Image analysis: use computer vision techniques to find features of images
 - Statistical inference
 - Descriptive statistics: summaries of your data, e.g., mean and variance of your variables
 - Analytic statistics: relationship between variables, studied through regressions, e.g., how are the number of faces in an instagram post associated with the likes it gets (Bakshi et al.,)
 - Causal inference: go beyond correlation, but often hard with observational social media data (more on this later in the course)
- But first: you have to make sure you have the right dataset

What is data cleaning and why do we do it?

- **Data cleaning** is the process of detecting and correcting or removing corrupt, inaccurate, or irrelevant records from a dataset to improve its quality before analysis.
- also called data processing, data preprocessing, data wrangling or data munging



Figure 1. Standard social media data cleaning pipeline.

https://htmlpreview.github.io/?https://github.com/dess-mannheim/SMDA/blob/main/Handouts/social_media_data_cleaning.html

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- Deduplication: Near-duplicates (e.g., retweets with slight edits) require fuzzy matching.

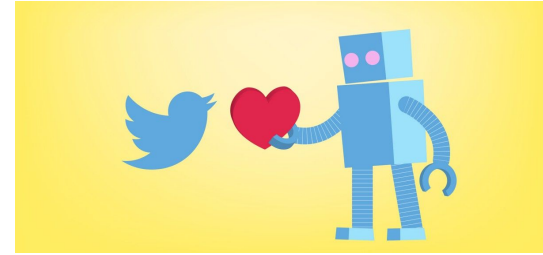


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Filtering Example 1: removing bots

- Bots are ubiquitous on many social media platforms
 - Some are good bots and some are evil
 - Some bots disclose they are bots (as many platform mandate them to)
 - Others impersonate real people to manipulate platform users (see next week's second reading)
- If your study doesn't focus on bots, ideally you would remove them from your sample
- Many tools exist for that, e.g., [Botometer](#)
- But, ascertaining whether a user is a bot or not is difficult, also for people!



Disagree? You Must be a Bot! How Beliefs Shape Twitter Profile Perceptions

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ABSTRACT

In this paper, we investigate the human ability to distinguish political social bots from humans on Twitter. Following motivated reasoning theory from social and cognitive psychology, our central hypothesis is that especially those accounts which are opinion-incongruent are perceived as social bot accounts when the account is ambiguous about its nature. We also hypothesize that credibility ratings mediate this relationship. We asked $N = 151$ participants to evaluate 24 Twitter accounts and decide whether the accounts were

1 INTRODUCTION

Today's social media landscape not only enables individuals to access a rich well of information and to connect with others but also allows for user-generated content to be published online. However, not all activity on social media platforms originates from human users. In recent years, concerns about the influence of automated accounts, so-called social bots, has risen. Social bots on social media are generally defined as automatic or semi-automatic accounts run by computer algorithms [18]. As they often mimic human behavior

Wischniewski, Magdalena, et al. "[Disagree? You must be a bot! How beliefs shape Twitter profile perceptions.](#)" *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 2021.

Filtering Example 2: removing irrelevant content

- Sometimes our data collection strategy can lead to the inclusion of irrelevant units
- Let's say you want to study presidential approval of Donald Trump
- You collect data with keywords like "Trump"
 - 'Trump' phrases unrelated to Donald Trump, which e.g. refer to Melania Trump or Donald Trump Jr.
- You have to now remove these irrelevant units
 - E.g., using a machine learning classifier or based on some rules (should also include the keyword 'president')
 - This cleaning technique will rarely be perfect



After Filtering: Normalizing data

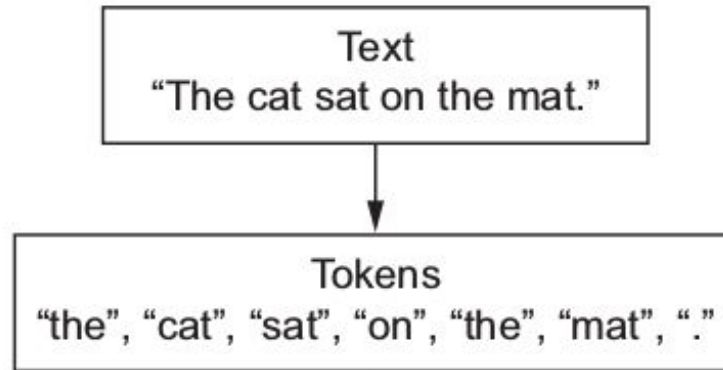
Element	Typical treatment
URLs	Replace with [URL] token or remove
Mentions (@user)	Replace with [USER] or remove
Hashtags (#topic)	Segment and keep text; remove #
Emojis	Convert to text description or keep as-is
HTML entities	Decode (& ; → &)
Lowercasing	Convert to lowercase
Punctuation	Task-dependent; often keep for sentiment

Text Data Cleaning

- tokenization
- remove stopwords
- Stemming / Lemmatization
- remove numbers
- remove headers and footers
- remove rare words
- Beyond Text:
 - dropping or imputing missing values
 - dropping columns with missing values

Text Data Cleaning

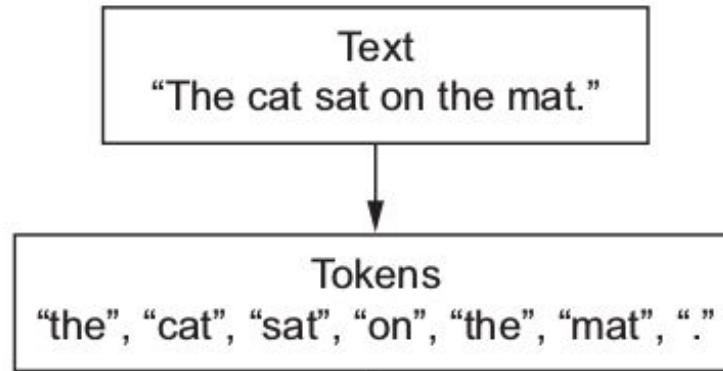
- **tokenization**
- remove stopwords
- Stemming ,
- remove nur
- remove hec
- remove rar
- Beyond Tex
 - dro
 - dro



Tokenization of a sentence

Text Data Cleaning

- **tokenization**: also digital trace data-based specific tokens such as @mentions, #hashtags, links, emojis, etc
- remove stopwords
- Stemming /
- remove non-ASCII characters
- remove HTML tags
- remove rare words
- Beyond Text Cleaning
 - document frequency
 - term frequency



Tokenization of a sentence

<https://github.com/jaredks/tweetokenize>

Text Data Cleaning

- tokenization
- remove stopwords
- **Stemming / Lemmatization**
- remove punctuation
- remove numbers
- remove special characters
- Beyond

Rule			Example		
SSSES	→	SS	caresses	→	caress
IES	→	I	ponies	→	poni
SS	→	SS	caress	→	caress
S	→		cats	→	cat

Examples of stemming

Case Study on data collection, cleaning, and analysis: Faces on Instagram posts

Studying Instagram

- **Construct:** how do the presence of faces, their gender and age impact engagement?
- **Data collection:** start with 2,000 popular posts and then snowball. Automatically infer faces, gender, and age using [Face++ API](#)
- **Data cleaning:** ??
- **Data analysis:** negative binomial regression which predicts the engagement of posts based on their content (faces, gender, and age of faces)
- **Result:** posts with faces have higher engagement, though age and gender do not have much effect

Faces Engage Us: Photos with Faces Attract More Likes and Comments on Instagram

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ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous.

ABSTRACT

Photos are becoming prominent means of communication online. Despite photos' pervasive presence in social media and online world, we know little about how people interact and engage with their content. Understanding how photo content might signify engagement, can impact both science and design, influencing production and distribution. One common type of photo content that is shared on social media, is the photos of people. From studies of offline behavior, we know that human faces are powerful channels of non-verbal communication. In this paper, we study this behavioral phenomena online. We ask how presence of a face, its age and gender might impact social engagement on the photo. We use a corpus of 1 million Instagram images and organize our study around two social engagement feedback factors, *likes* and *comments*. Our results show that photos with faces are 38% more likely to receive likes and 32% more likely to receive comments, even after controlling for social network reach and activity. We find, however, that the number of faces, their age and gender do not have an effect. This work presents the first results on how photos with human faces relate to engagement on large scale image sharing communities. In addition to contributing to the research around online user behavior, our findings offer a new line of future work using visual analysis.

Author Keywords

mobile; photo; Instagram; faces; face detection; engagement; age; demographics; gender; content; image; social media; image sharing community

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INTRODUCTION

Even as babies, humans love to look at faces; infants, barely minutes old, turn toward faces, sensing that they are important [39]. It is widely accepted in neuroscience that face perception is perhaps the most highly developed human visual skill [26]. Faces are also powerful channels of nonverbal communication [45]. We constantly monitor faces because they provide vital clues in an impressive variety of contexts: attraction, the complexity of emotions, identity, age, humor, and a person's regional and national background [24].

Many of the faces we see everyday now have an online presence. Photo sharing communities such as Instagram have made it possible to communicate with large groups of distributed people through an image—be it a picture of what's for dinner or a selfie—perhaps more easily than through words alone. As Kelsey puts it, “we are moving away from photography as a way of recording and storing the past, and instead turning photography into a social medium in its own right” [35].

Online photo sharing communities have grown at an impressive pace. At the time of this writing, Instagram users upload 55 million photos a day to the site¹. This presents a key research challenge for photo sharing communities like Instagram (cf. Flickr, Imgur, Tumblr): how do we discover the mechanisms by which users communicate around visual content and engage with such content. In other words, since engagement is vital to photo sharing communities, it is critical to understand what form of content drives engagement. While several research studies have focused on how users engage with textual content [6, 9, 11, 12, 31, 38], there are few studies on what makes visual content socially engaging online. To investigate this, we ask the following research questions in this paper, driven by social psychology work on faces

Bakhshi, Saeideh, David A. Shamma, and Eric Gilbert. [“Faces engage us: Photos with faces attract more likes and comments on instagram.”](#)

Proceedings of the SIGCHI conference on human factors in computing systems. 2014.

Studying Instagram

- **Construct:** how do the presence of faces, their gender and age impact engagement?
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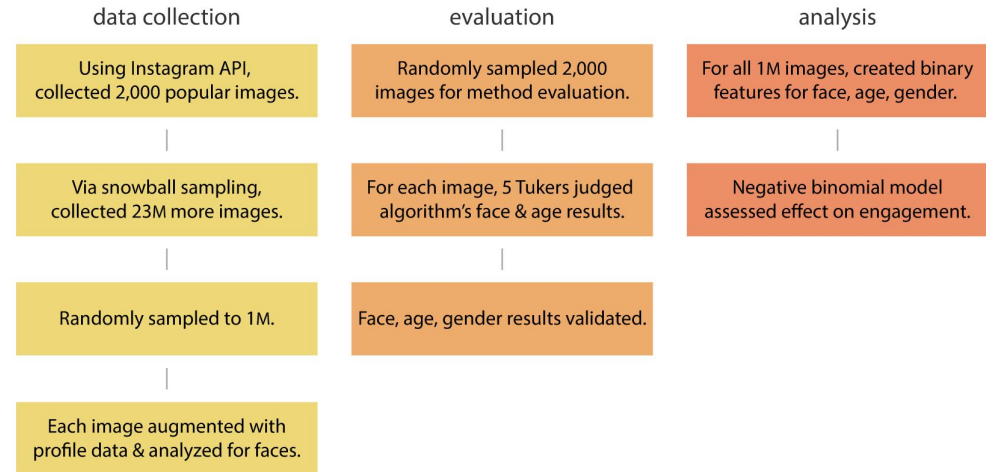


Figure 2. An overview of the steps taken in this paper for collection, evaluation and analysis of the data.

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- **Result:** posts with faces attract more likes and comments, though age and gender do not have much effect

Is this data collection, processing, or analysis?

Faces Engage Us: Photos with Faces Attract More Likes and Comments on Instagram

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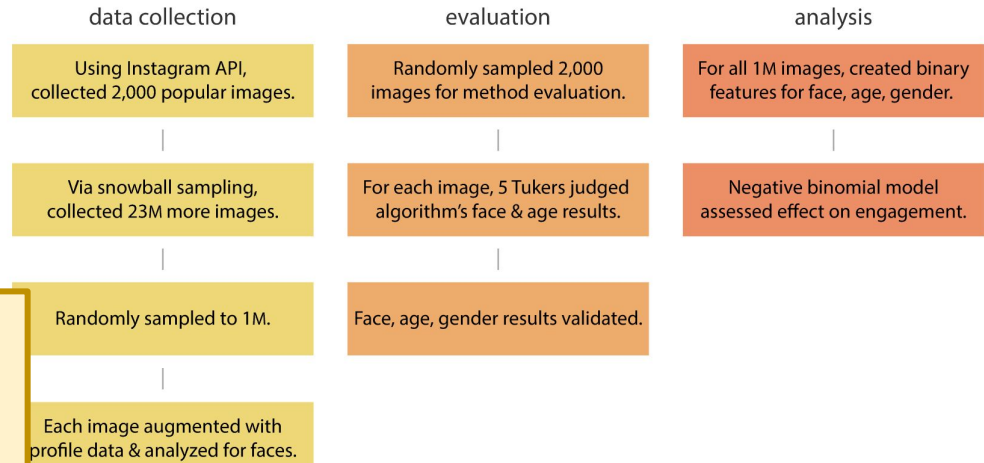


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Data Analysis vs. Collection/Processing

- sometimes, there isn't a clear separation between the steps; collection, processing, and analysis may go hand in hand
- Data augmentation can sometimes be a part of data collection or data cleaning — more of a stylistic choice
- steps may range from text cleaning (parsing the data, removing stopwords) to complex analysis (create text networks or get stance of the post)

Data Cleaning has Consequences

Element	Typical treatment	Caution
URLs	Replace with <code>[URL]</code> token or remove	URL domain can be a signal (news vs. spam)
Mentions (<code>@user</code>)	Replace with <code>[USER]</code> or remove	Preserve for network analysis
Hashtags (<code>#topic</code>)	Segment and keep text; remove <code>#</code>	Often contain key semantic content
Emojis	Convert to text description or keep as-is	Emojis are critical for sentiment tasks
HTML entities	Decode (<code>&amp;</code> → <code>&</code>)	Always decode before any analysis
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Therefore, we should always

- Validate data cleaning steps
- Document data cleaning steps

Today:

1. Data Cleaning
- 2. Studying Wikipedia**

Why Wikipedia?

- Wikipedia is essential for the online information ecosystem
- Used for fact-checking
- Used to train large language models

The Most Important Laboratory for Social Scientific and Computing Research in History

Benjamin Mako Hill

Aaron Shaw

October 24, 2019*

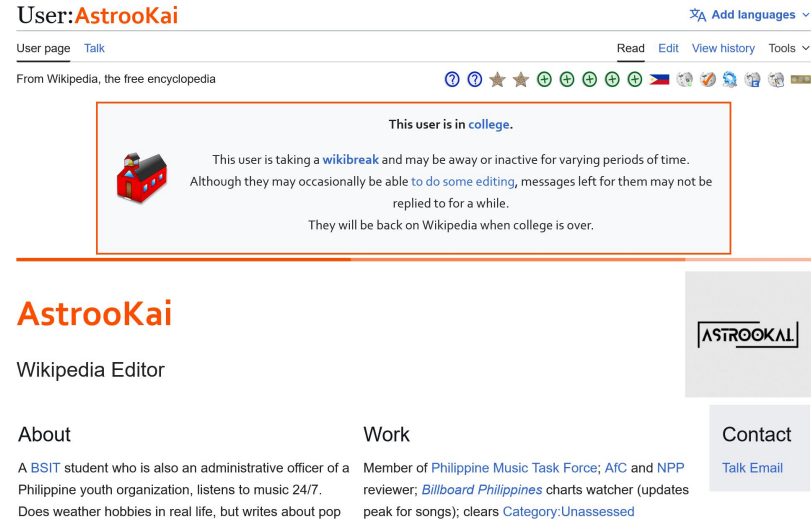
Twenty years ago, Benjamin Mako Hill, Benjamin Mako, and Aaron Shaw. "The most important laboratory for social scientific and computing research in history." *Reagle*. Results from Google Scholar (2020) that over 6,000 scholarly publications mention Wikipedia in their title and over 1,700,000 mention it somewhere in their text. For comparison, the phrase "Catholic Church"—an organization with a nearly 2,000 year head start—returns about the same number of mentions in publication titles. In under twenty years, Wikipedia has become one of the most heavily studied organizations of any kind. To the extent that Wikipedia research is a field of study, what major areas of investigation have been pursued in the field so far? What are the big discoveries? The most striking gaps? This essay addresses these questions and considers some of the most important directions

Project(s) ◆	Articles ◆	All pages ◆	Edits ◆	Admins ◆	Users ◆	Active users ◆
English Wikipedia	7,153,199	65,303,242	1,339,190,147	814	51,964,146	283,122
All active Wikipedias	66,761,628	266,276,982	3,849,866,401	3,405	128,067,287	630,690

<https://en.wikipedia.org/wiki/Wikipedia:Statistics>

A little about Wikipedia

- Wikipedia as a social platform
 - The articles contain information about society
 - Editors communicate with each other while editing articles
 - Edits wars
 - Vandalism
 - Sockpuppets
- Wikipedia is operated by the Wikimedia Foundation
 - It is a non-profit organization of ~700 staff
 - It provides broad support to Wikimedia communities and
 - projects: servers, data centers, legal and communications, support, etc.
 - It does not create or modify content.
 - It does not define or enforce policies on the projects



The screenshot shows the Wikipedia user page for 'User:AstrooKai'. At the top, it says 'User:AstrooKai' with a language dropdown. Below that are links for 'User page' and 'Talk'. A banner at the top right says 'From Wikipedia, the free encyclopedia'. A central box contains a red house icon and text stating: 'This user is in college. This user is taking a **wikibreak** and may be away or inactive for varying periods of time. Although they may occasionally be able to **do some editing**, messages left for them may not be replied to for a while. They will be back on Wikipedia when college is over.' Below this, the username 'AstrooKai' is displayed in large orange letters, followed by 'Wikipedia Editor'. To the right is a grey box with 'ASTROOKAI' in a stylized font. Below the username, there are sections for 'About' and 'Work'. The 'About' section describes the user as a BSIT student and administrative officer. The 'Work' section lists their involvement with the Philippine Music Task Force and NPP, as well as their role as a Billboard Philippines charts watcher. A 'Contact' box with 'Talk Email' links is on the right.

User:AstrooKai

User page Talk

Read Edit View history Tools

From Wikipedia, the free encyclopedia

This user is in college.

This user is taking a **wikibreak** and may be away or inactive for varying periods of time. Although they may occasionally be able to **do some editing**, messages left for them may not be replied to for a while. They will be back on Wikipedia when college is over.

AstrooKai

Wikipedia Editor

About

A BSIT student who is also an administrative officer of a Philippine youth organization, listens to music 24/7. Does weather hobbies in real life, but writes about pop

Work

Member of **Philippine Music Task Force**; AIC and NPP reviewer; **Billboard Philippines** charts watcher (updates peak for songs); clears **Category:Unassessed**

Contact

Talk Email

<https://en.wikipedia.org/wiki/User:AstrooKai>

Source: https://meta.wikimedia.org/wiki/Wikimedia_Data_Tutorial_ICWSM_2024

What makes Wikipedia different from other Social Media Platforms?

- Open access
 - Commitment to Open Access Policy <https://w.wiki/7APO>
- Wikimedia Research Team aims to make any work produced with our support openly available to the public and reusable on Wikimedia projects.
 - Open licenses
 - Content can be freely re-used <https://w.wiki/8Bur>
 - e.g. CC-BY-SA 4.0
- Transparency
 - Data pipelines are public
 - Example: clickstream dataset <https://w.wiki/ACVn>
- Privacy
 - No 3rd-party collection or sharing of user data <https://w.wiki/6awx>
 - Private user data removed after 90 days

Let's look at some studies that use Wikipedia data

Automatic Vandalism Detection in Wikipedia

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Abstract. We present results of a new approach to detect destructive article revisions, so-called vandalism, in Wikipedia. Vandalism detection is a one-class classification problem, where vandalism edits are the target to be identified among all revisions. Interestingly, vandalism detection has not been addressed in the Information Retrieval literature by now. In this paper we discuss the characteristics of vandalism as humans recognize it and develop features to render vandalism detection as a machine learning task. We compiled a large number of vandalism edits in a corpus, which allows for the comparison of existing and new detection approaches. Using logistic regression we achieve 83% precision at 77% recall with our model. Compared to the rule-based methods that are currently applied in Wikipedia, our approach increases the F -Measure performance by 49% while being faster at the same time.

Introduction. The content of the well-known Web encyclopedia Wikipedia is created collaboratively by volunteers. Every visitor of a Wikipedia Web site can participate immediately in the authoring process: articles are created, edited, or deleted without need for authentication. In practice, an article is developed incrementally since, ideally, authors review and revise the work of others. Till this day about 8 million articles in 253 languages have been authored in this way.

Potthast, Martin, Benno Stein, and Robert Gerling. "Automatic vandalism detection in Wikipedia." *European conference on information retrieval*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2008.

Rama et al. *EPJ Data Science* (2022) 11:1
<https://doi.org/10.1140/epjds/s13688-021-00312-8>



REGULAR ARTICLE

Open Access



A large scale study of reader interactions with images on Wikipedia

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Abstract

Wikipedia is the largest source of free encyclopedic knowledge and one of the most visited sites on the Web. To increase reader understanding of the article, Wikipedia editors add images within the text of the article's body. However, despite their widespread usage on web platforms and the huge volume of visual content on Wikipedia, little is known about the importance of images in the context of free knowledge environments. To bridge this gap, we collect data about English Wikipedia reader interactions with images during one month and perform the first large-scale analysis of how interactions with images happen on Wikipedia. First, we quantify the overall engagement with images, finding that one in 29 pageviews results in a click on at least one image, one order of magnitude higher than interactions with other types of article content. Second, we study what factors associate with image engagement and observe that clicks on images occur more often in shorter articles and articles about visual arts or transports and biographies of less well-known people. Third, we look at interactions with Wikipedia article previews and find that images help support reader information need when navigating through the site, especially for more popular pages. The findings in this study deepen our understanding of the role of images for free knowledge and provide a guide for Wikipedia editors and web user communities to enrich the world's largest source of encyclopedic knowledge.

Keywords: Wikipedia; Images; Computer vision; User behavior

Rama, Daniele, et al. "A large scale study of reader interactions with images on Wikipedia." *EPJ Data Science* 11.1 (2022): 1.

Let's look at some studies that use Wikipedia data

Sudden Attention Shifts on Wikipedia During the COVID-19 Crisis

Why We Read Wikipedia

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ABSTRACT

Wikipedia is one of the most popular sites on the Web, with millions of users relying on it to satisfy a broad range of information needs every day. Although it is crucial to understand what exactly these needs are in order to be able to meet them, little is currently known about why users visit Wikipedia. The goal of this paper is to fill this gap by combining a survey of Wikipedia readers with a log-based analysis of user activity. Based on an initial series of user surveys, we build a taxonomy of Wikipedia use cases along several dimensions, capturing users' motivations to visit Wikipedia, the depth of knowledge they are seeking, and their knowledge of the topic of interest prior to visiting Wikipedia. Then, we quantify the prevalence of these use cases via a large-scale user survey conducted on live Wikipedia with almost 30,000 responses. Our analyses highlight the variety of factors driving users to Wikipedia, such as current events, media coverage of a topic, personal curiosity, work or school assignments, or boredom. Finally, we match survey responses to the respondents' digital traces in Wikipedia's server logs, enabling the discovery of behavioral patterns associated with specific use cases. For instance, we observe long and fast-paced page sequences across topics for users who are bored or exploring

per day. It attracts millions of readers from across the globe and serves a broad range of their daily information needs. Despite this, very little is known about the motivations and needs of this diverse user group: why they come to Wikipedia, how they consume the content in the encyclopedia, and how they learn. Without this knowledge, creating more content, products, and services that ensure high levels of user experience remains an open challenge [3, 8, 20, 43].

Background and objectives. A rich body of work has investigated motivations and behavior patterns of users on the Web [11, 21]. Specific attention has been cast on a few major sites including search engines [6, 33, 41], and social networking sites such as Twitter [15, 22] and Facebook [34]. Yet, surprisingly little is known about the motivations, needs, and behaviors of Wikipedia readers, possibly keeping Wikipedia from reaching its full potential.

The vast literature on user behavior in Wikipedia (*cf.* Okoli *et al.* [30] for an overview) has focused on content production. It mainly investigates editors' motivations [1, 29], patterns of editing behavior [16], and the quality of content [18, 39]. Much less is known about content consumption, even though readers make up the majority of Wikipedia users. The limited work on readers has focused on topics such as content preferences [26, 32, 37], search queries leading to Wikipedia

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Abstract

We study how the COVID-19 pandemic, alongside the severe mobility restrictions that ensued, has impacted information access on Wikipedia, the world's largest online encyclopedia. A longitudinal analysis that combines pageview statistics for 12 Wikipedia language editions with mobility reports published by Apple and Google reveals massive shifts in the volume and nature of information seeking patterns during the pandemic. Interestingly, while we observe a transient increase in Wikipedia's pageview volume following mobility restrictions, the nature of information sought was impacted more permanently. These changes are most pronounced for language editions associated with countries where the most severe mobility restrictions were implemented. We also find that articles belonging to different topics behaved differently; e.g., attention towards entertainment-related topics is lingering and even increasing, while the interest in health- and biology-related topics was either small or transient. Our results highlight the utility of Wikipedia for studying how the pandemic is affecting people's needs, interests, and concerns.

1 Introduction

The coronavirus disease 2019 (COVID-19) pandemic has led to the implementation of unprecedented non-pharmaceutical interventions ranging from case isolation to national lockdowns. Ribeiro, Manoel Horta, et al. "Sudden attention shifts on Wikipedia following COVID-19 mobility restrictions." *Proceedings of the*

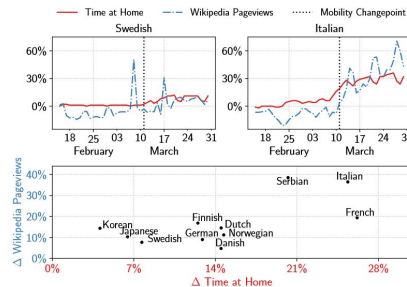


Figure 1: Wikipedia access vs. mobility — Association between increase in time spent at home (from Google mobility reports; red) and increase in Wikipedia access volume (from Wikipedia pageview statistics; blue), both in terms of relative change over a five-week baseline period in early 2020. Top: time series for Sweden/Swedish Wikipedia and Italy/Italian Wikipedia; dotted vertical lines: changepoints in mobility time series. Bottom: summary for 11 of the 12 languages. x-axis: increase in time at home; y-axis: increase in total Wikipedia pageviews.

Singer, Philipp, et al. "Why we read Wikipedia." *Proceedings of the 26th international conference on world wide web*. 2017.

International AAAI Conference on Web and Social Media. Vol. 15. 2021.

Let's revisit the Wikipedia Gender Bias Study

- **Research Question:** Is there gender bias in Wikipedia articles about women?
- **Construct:** gender bias
- **Platform:** Wikipedia
- **Data Collection Strategy:** start with an external list of notable people and then match with Wikipedia
- **Data Analysis Strategy:**
 - statistical analysis of coverage
 - lexical analysis of how women's and men's articles differ
 - network analysis of articles to see links to articles
- **Results:** women and men have similar coverage, but women are described differently (in relation to other men)

It's a Man's Wikipedia? Assessing Gender Inequality in an Online Encyclopedia

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Abstract

Wikipedia is a community-created encyclopedia that contains information about notable people from different countries, epochs and disciplines and aims to document the world's knowledge from a neutral point of view. However, the narrow diversity of the Wikipedia editor community has the potential to introduce systemic biases such as gender biases into the content of Wikipedia. In this paper we aim to tackle a sub problem of this larger challenge by presenting and applying a computational method for *assessing gender bias on Wikipedia along multiple dimensions*. We find that while women on Wikipedia are covered and featured well in many Wikipedia language editions, *the way women are portrayed* starkly differs from the way men are portrayed. We hope our work contributes to increasing awareness about gender biases online, and in particular to raising attention to the different levels in which gender biases can manifest themselves on the web.

Introduction

Wikipedia aims to provide a platform to freely share the sum of all human knowledge. It represents an influential source of information on the web, containing encyclopedic information about notable people from different

of gender biases on Wikipedia. Yet, due to the influential nature of Wikipedia, it is important to reveal, assess and correct such biases, if they exist. This paper tackles a sub-part of this larger challenge.

Objectives: In particular, the overall goal of this work is to *assess potential gender inequalities in Wikipedia articles* along different dimensions.

Approach: To assess the extent to which Wikipedia suffers from potential gender bias, we analyze articles about notable people in six language editions along four different gender bias dimensions: coverage bias, structural bias, lexical bias and visibility bias. *Coverage bias* determines differences between the number of notable women and men portrayed on Wikipedia. For example, one might hypothesize that notable men are more likely to be covered by Wikipedia. *Structural bias* quantifies gender homophily/disassortativity, i.e. gender-specific tendencies to preferably link articles of notable people with the same or different gender. For example, one might hypothesize that articles about women have more links to men than vice versa. *Lexical bias* reveals inequalities in the words used to describe notable men and women on Wikipedia. For example, articles about women

Wagner, Claudia, et al. "[It's a man's Wikipedia? Assessing gender inequality in an online encyclopedia](#)." *Proceedings of the international AAAI conference on web and social media*. Vol. 9. No. 1. 2015.

own gender bias in the population of editors has led to introduce gender biases into the contents of Wikipedia as well. For example, the population bias might lead to differences in the ways women and men are portrayed on Wikipedia. It might also mimic or even exaggerate inequalities that are already existing in the real world. At the same time, assessing the manifold and subtle ways in

mention their family (husband or wife) to the front page of the article. *Visibility bias* reflects how often an article is selected.

present and apply a computational method for *assessing gender bias on Wikipedia along multiple dimensions*. We find that most Wikipedia language editions exhibit a slight over-representation of women, but the proportional differences in the coverage of men and women are not significant. That means, men and women are covered equally well in all six Wikipedia language editions. Also on the visibility level, we do not find

Wikipedia Gender Bias Study: Data Processing

- **Data Analysis Strategy:**
 - lexical analysis of how women's and men's articles differ
- Build a classifier which finds words in an article that can distinguish the gender of the article's subject
- The words are then categorized into:
 - Gender (i.e., man, women, mrs, mrs, lady, gentleman)
 - Relationship (e.g., married, divorced, couple, husband, wife)
 - Family (e.g., kids, children, mother, grandmother).

It's a Man's Wikipedia?
Assessing Gender Inequality in an Online Encyclopedia

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Lexical Bias: To explore gender-specific lexical inequalities on Wikipedia we use an open vocabulary approach, inspired by (Schwartz et al. 2013). An open-vocabulary approach is not limited to predefined word lists, but linguistics are automatically determined from the text. We compute the tfidf scores of the word stems obtained from a Snowball Stemmer and use them as features to train a Naive Bayes classifier. The classifier determines which words are most effective in distinguishing the gender of the person an article is about. Log likelihood ratios $L(word, g)$ are used for comparing different feature-outcome relationships.

$$L(word, g) = \log \left(\frac{P(word|g)}{P(word)} \right) \quad (3)$$

where $P(word|g)$ is the conditional distribution that a word shows up in an article about a person given that the person's gender is g , and $P(word)$ is the probability that a word shows up in any article regardless of the gender of the person the article is about.

Wagner, Claudia, et al. "It's a man's Wikipedia? Assessing gender inequality in an online encyclopedia." *Proceedings of the international AAAI conference on web and social media*. Vol. 9. No. 1. 2015. Wikipedia. It might also mimic or even exaggerate the characteristics that are already existing in the real world. At the same time, assessing the manifold and subtle ways in which gender inequality is represented on Wikipedia. Again, one can hypothesize that articles about women are disproportionately over-represented in the coverage of men and women are not significant. That means, men and women are covered equally well in all six Wikipedia language editions. Also on the visibility level, we do not find

- **Data Analysis Strategy:**

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Figure 8: **Lexical Bias:** The proportion of the 150 most discriminative words of articles about women that belong to different categories. In all language editions between 32% and 23% of the 150 most indicative words for women belong to one of the three categories, while only between 0% and 4% of the most discriminative words for men belong to one of these categories. In some language edition, like the Russian (RU), the English (EN) and the German (DE) one, the proportion of the most discriminative words that belong to one of these three categories is especially high among the top words.

Wagner, Claudia, et al. "It's a man's Wikipedia? Assessing gender inequality in an online encyclopedia." *Proceedings of the international AAAI conference on web and social media*. Vol. 9. No. 1. 2015. Wikipedia. It might also mimic or even exaggerate what are already existing in the real world. At the same time, assessing the manifold and subtle ways in which gender inequality is reflected in Wikipedia. Again, one can hypothesize that articles about women, but one prominent difference in the coverage of men and women are not significant. That means, men and women are covered equally well in all six Wikipedia language editions. Also on the visibility level, we do not find

Discuss: Data Collection and Cleaning Decisions

- Choose one of the following use cases:
 - Studying the factuality of nutrition-related posts on Instagram
 - Studying the factuality of movie-related posts on Wikipedia
- How would you collect data for the chosen use case?
 - Could there be any data quality problems with your approach?
- How would you clean the data?
 - Could there be any data quality problems with your approach?

Summary and Next Week

- This session we learned:
 - Cleaning and Preprocessing Social Media Data
 - Consequences of Cleaning
 - Wikipedia analysis
- Next time:
 - Exploratory Data Analysis
 - Word Clouds and other related visualizations
 - Readings:
 - Gallagher, Ryan. J., Morgan R. Frank, Lewis Mitchell, Aaron J. Schwartz, Andrew J. Reagan, Christopher M. Danforth, and Peter Sheridan Dodds. "[Generalized Word Shift Graphs: A Method for Visualizing and Explaining Pairwise Comparisons Between Texts](#)." EPJ Data Science 10, no. 4 (2021).
 - Nizzoli, Leonardo, et al. "[Coordinated behavior on social media in 2019 UK general election](#)." Proceedings of the international AAI conference on web and social media. Vol. 15. 2021.

Questions?

Additional Resources

- Tutorial on Wikipedia data collection and analysis: Martin Gerlach, Lucie-Aimee Kaffee, Tiziano Piccardi:
https://meta.wikimedia.org/wiki/Wikimedia_Data_Tutorial_ICWS_M_2024